Policy in Action Project - Final presentation

17/03/2026

Environmental Impact Assessment of the DGFiP Gen AI Platform

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GENERAL FRAMEWORK

- We assessed the direct and indirect environmental impacts of AI with a **literature review for all the stages of the platform's lifecycle.**

- **Environmental Life-Cycle Assessment (LCA):** go beyond operational electricity.

(Berthelot et al., 2024; Plociennik et al., 2025; Ciambrone 1997)

(Luccioni et al., 2023)

- **Key output:** CO_2 equivalent

GENERAL FRAMEWORK

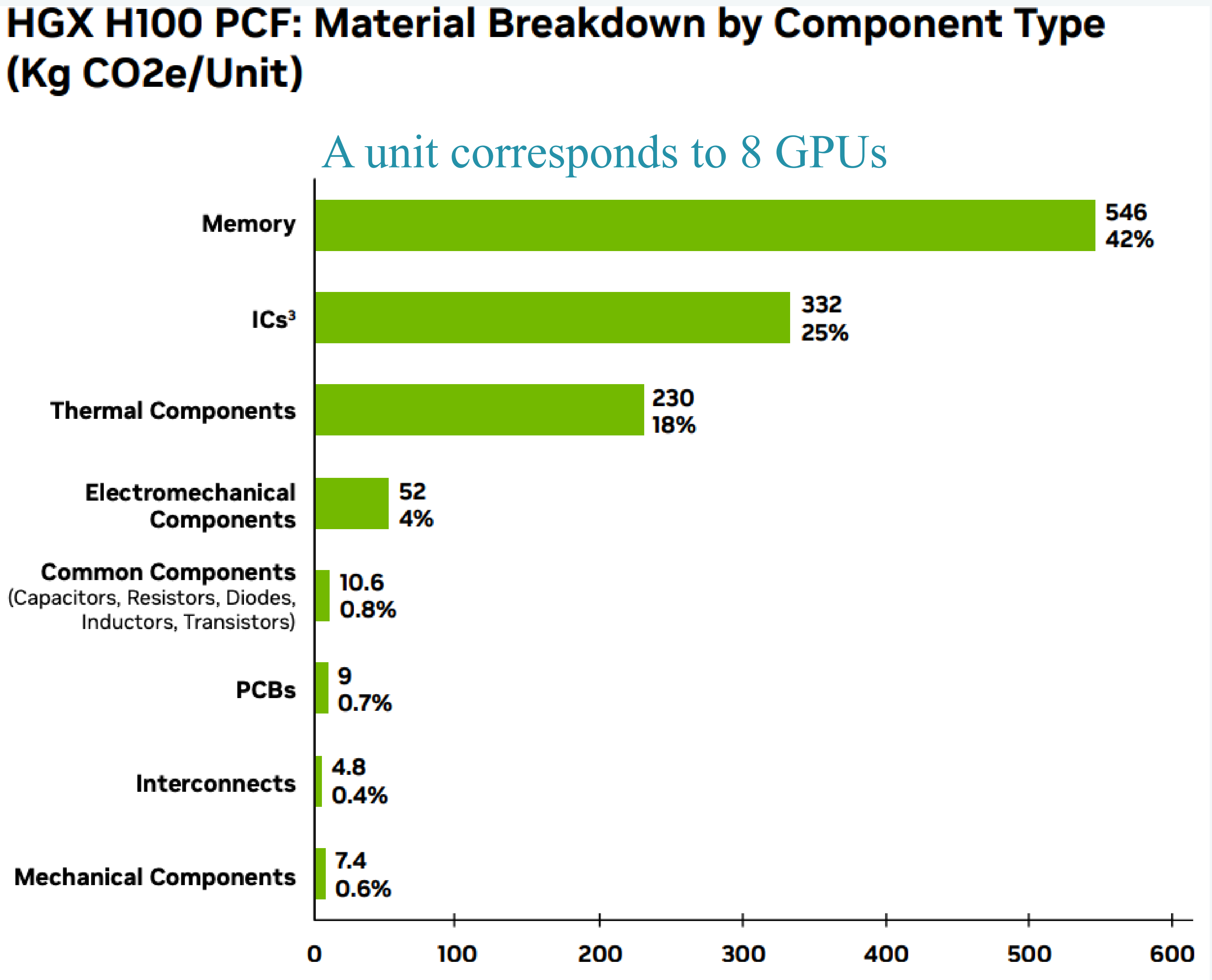
- **Different scopes of emissions :** (Luccioni, Viguier, and Ligozat 2023)
  - **On-site emissions:** Diesel back-up generators, leakage of refrigerant gases => 2% of total life cycle emissions
  - **Direct energy emissions:** Consumption of local servers from inference => ~ 70% of total life cycle emissions
  - **Embodied emissions from the value chain:** Manufacturing, transportation, end of life of hardware, and a fraction of the initial training (training debt) => 20% to 30% of the total life cycle emissions

=> The percentages depend on the energy grid, size of models, hardware used, lifetime of model ...

PHASE I: HARDWARE AND INFRASTRUCTURE

- Environmental impacts of the AI hardware
- **Scope:** “Cradle-to-Gate” assessment: from raw material extraction to the exit of the factory
- **Breakdown:**
  - **Components:** 91% of the footprint (Memory, ICs, ...)
  - **Assembly and transport:** 9% of the footprint.
- **Baseline:** 1 standard HGX baseboard (8 GPUs) = 1,312 kg CO2e
- **DGFiP fleet (24 GPUs / 3 baseboards):** 3,936 kg CO2e (3.9t)
- **Equivalencies:**
  - **18,000 km** driven by a thermal car.
  - 15 full computers (excluding screens).

  (ADEME, impact CO2 equivalent calculator (2025))



(NVIDIA's study of the carbon footprint of their HGX H100 80GB HBM3 GPU)

Key Insight: GPUs represents a relatively small part of the total lifecycle emissions, especially when amortized over a 4-year usage period.

PHASE I: HARDWARE AND INFRASTRUCTURE

- **Beyond CO2 emissions**, a single advanced GPU manufacturing process can consume over **30,000 liters of water**.

- We also need to take into account the emissions from the data center hosting the GPUs infrastructure.

- **PUE (Power Usage Effectiveness):**
  - **Current DGFiP PUE 1.7:** For every 1 kWh used by GPUs, 0.7 kWh is wasted on support (cooling/lighting) so the total demand is 1.7 kWh.
  - **Target** of 1.4 PUE

- Other CO2 emissions not taking into account:
  - **Refrigerants gases leakage:** R-134a has a GWP (Global Warming Potential) 1,430x higher than CO2
  - **Back-up power:** Diesel generators emit CO2 during monthly testing and power outages.

PHASE I: HARDWARE AND INFRASTRUCTURE

Strategic recommendations for DGFiP

- **Hardware longevity:** Keep GPUs to maximize the amortization of manufacturing emissions.
- **Change in hardware:** When hardware must be replaced, prioritize efficiency (e.g., NVIDIA B200 chips offer significantly better energy-per-token performance).
- **Cooling:** Transition to water cooling systems. It can reduce PUE to ~1.1, cutting the operational footprint by 30%.

PHASE II: INITIAL TRAINING

Even as a downstream user, **DGFiP inherits the environmental cost of every model it deploys**.

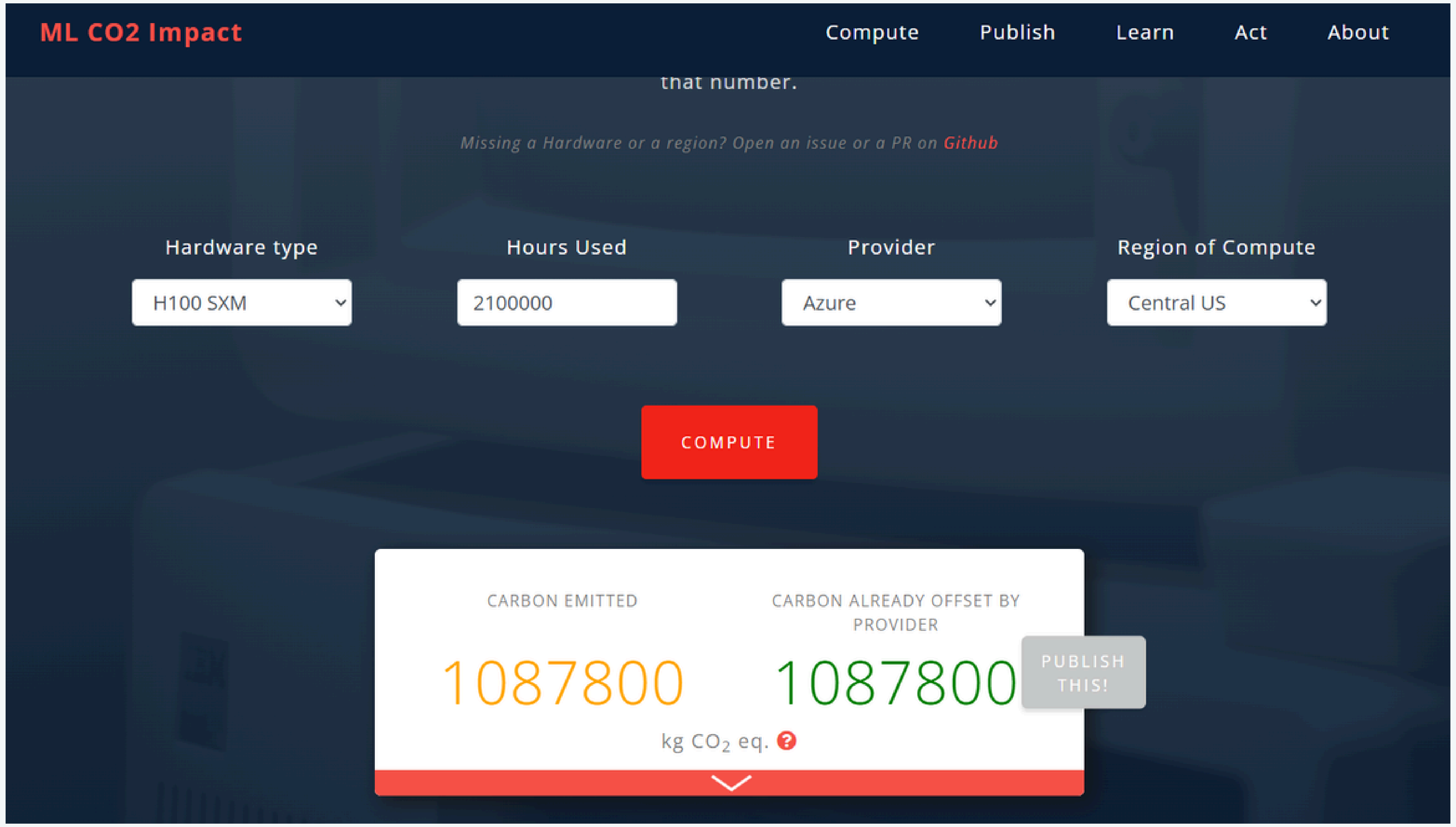
- Precise fractional attribution is impossible : **Total global inference volume for each model is unknown**, DGFiP can't calculate its exact share of training debt.
- **DGFiP should rely on total initial training emissions** as evaluation and selection criterias of models. Problem : 99% of models on Hugging Face do not report emissions data. (Castaño et al. 2023)

Model	Size	Location	Data Quality	Estimation tCO_2e
Llama-3.3-70B	70B	US	Reported	2040
GPT-OSS-120B	120B	US	Partial	≈ 1087
GPT-OSS-20B	21B	US	Partial	≈ 108,7
Qwen2.5-Coder-32B	32B	China	Partial	≈ 466,4
Mistral-Small-24B	24B	France/US	Unknown	—
Whisper-Large-v3	809M	Mixed	Unknown	—
gte-Qwen2-1.5B	1.5B	China	Unknown	—
dgfip-e5-large	335M	France	Unknown	—

Energy Awareness   Context Reporting   Unknown emissions

PHASE II: INITIAL TRAINING

Context Reporting

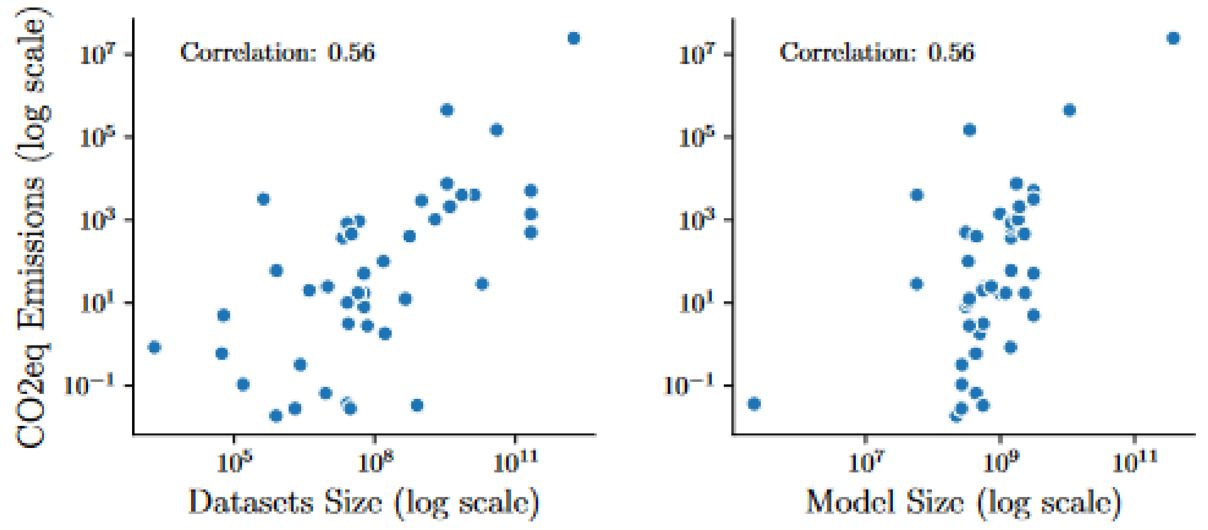


Machine Learning CO2 Impact Calculator, Green Algorithms Calculator

Unknown emissions

Key proxy indicators when data is missing:

- Dataset volume / training hours
- Hardware/PUE
- Training location
- Model size



Correlation between carbon emissions and dataset / model size

Ariyanti et al. 2025

PHASE II: INITIAL TRAINING

Strategic recommendations for DGFiP

- Prioritise models with **published Model Cards and carbon reporting.**
- If context reporting, use tools like **ML CO2 Impact** or **Green Algorithms** to approximate emissions.
- If no context, use proxies (dataset size, hardware, training location)
- **Favour smaller models**.

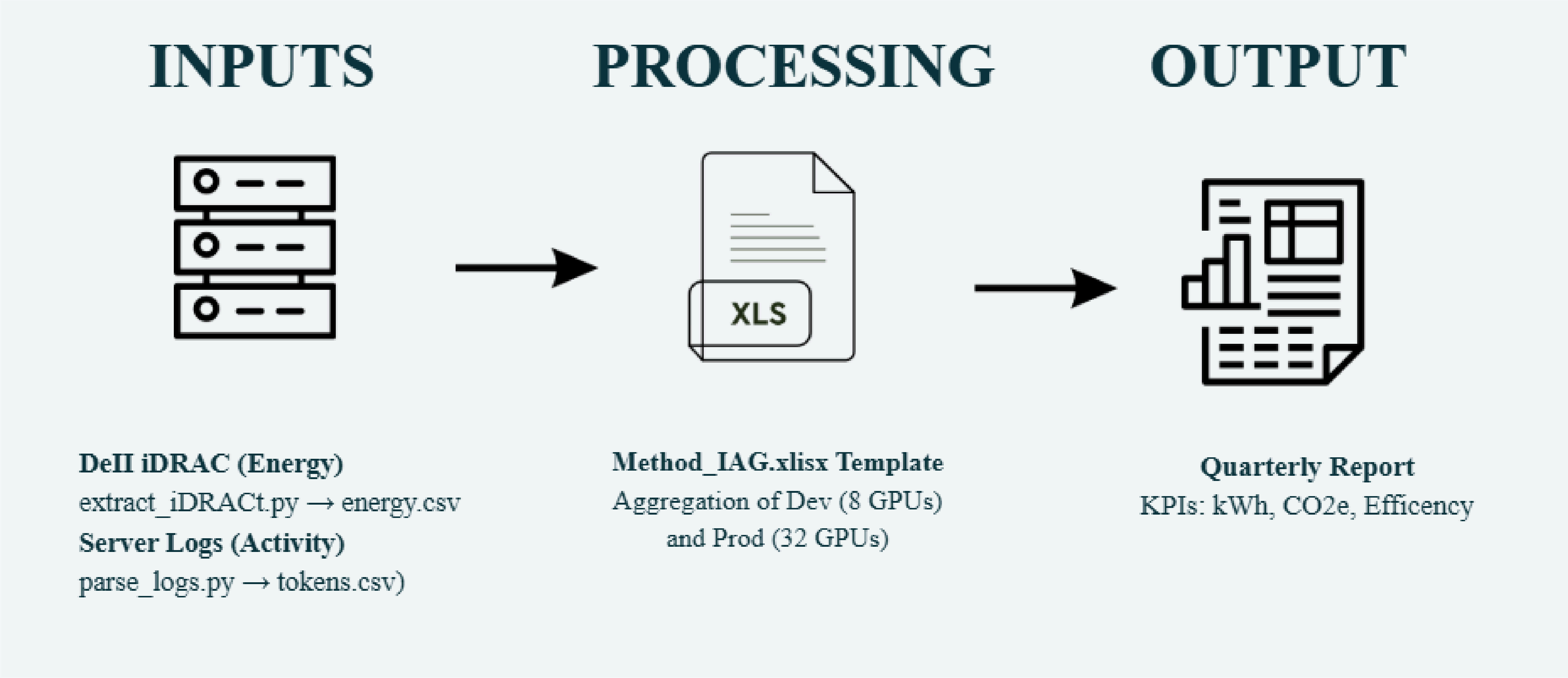
=> Higher training footprints typically also mean higher inference costs

PHASE III: INFERENCE
Assement tool of inference CO_2 emissions

- **CO_2 emission from inference:** Energy consumption (W) x PUE x electricity mix=> the tool will present in detail the methodology to calculate the CO_2 emissions of the inference from DGFiP models

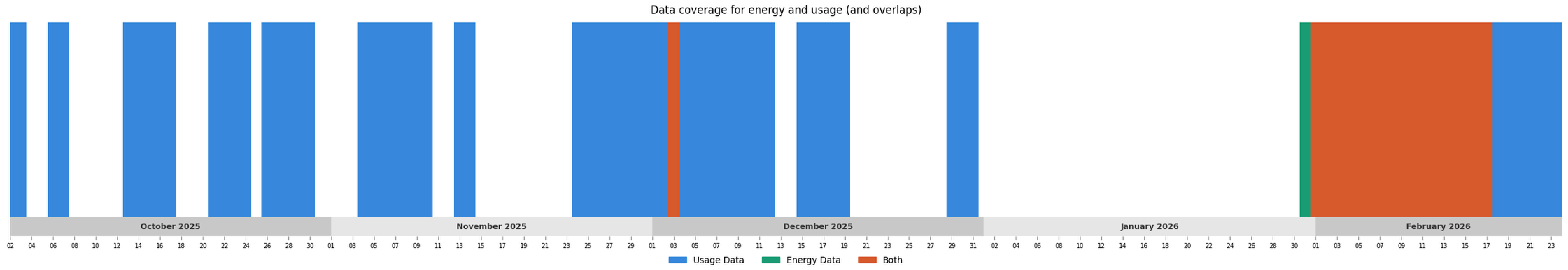
- **CO_2 emission from idleness (sleeping):** servers still consume **10% of peak power**. An NVIDIA H100 GPU draws 50–60W at 0% utilization. For a cluster of 24 GPUs, that is **1.2–1.4 kW of power, 24/7**.
  (NVIDIA carbon footprint of their HGX H100 GPU)

- **What you asked us to do:**

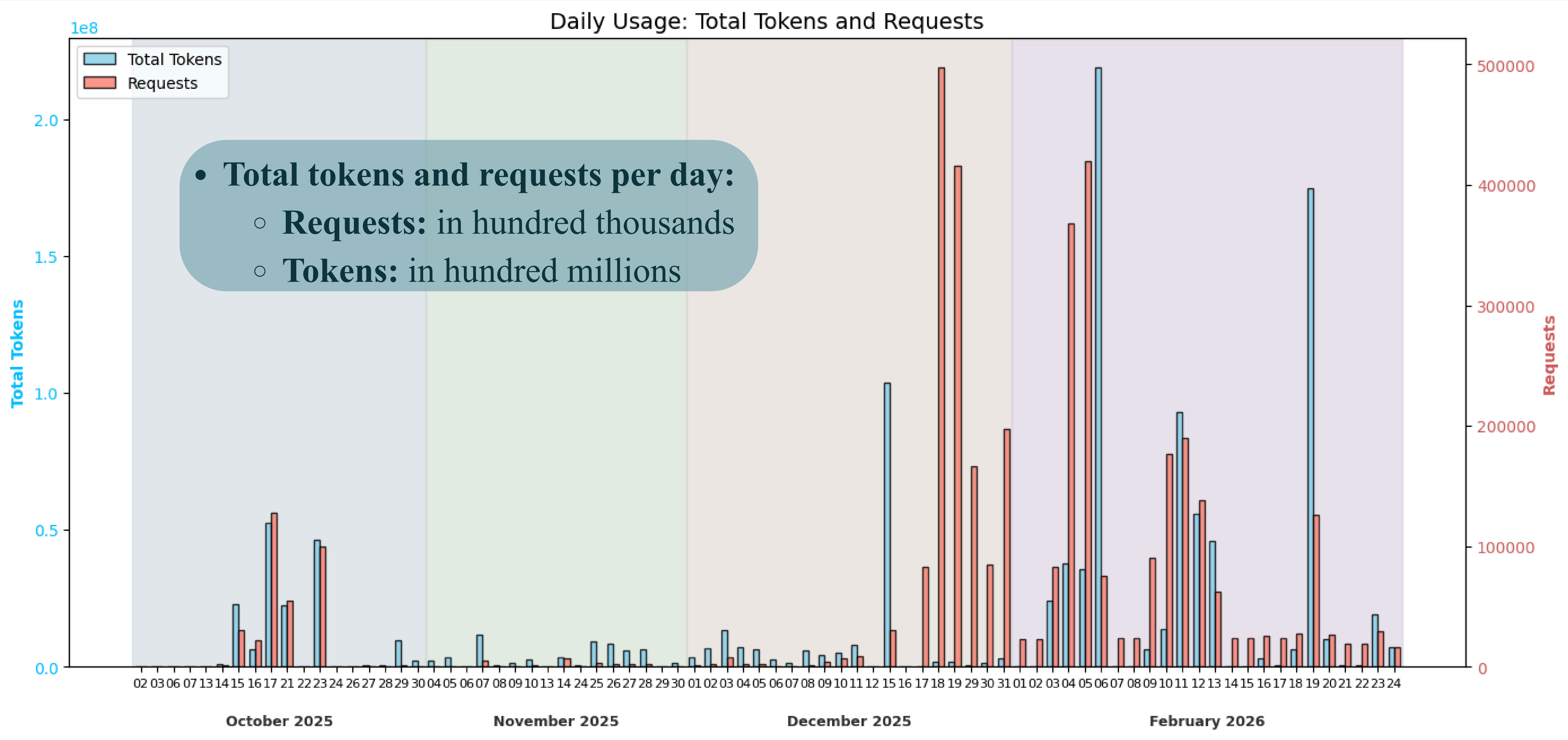


PHASE III: INFERENCE
Descriptive statistics: Data coverage

- **Data coverage we have:** only 17 days of overlap between the energy data and the usage data

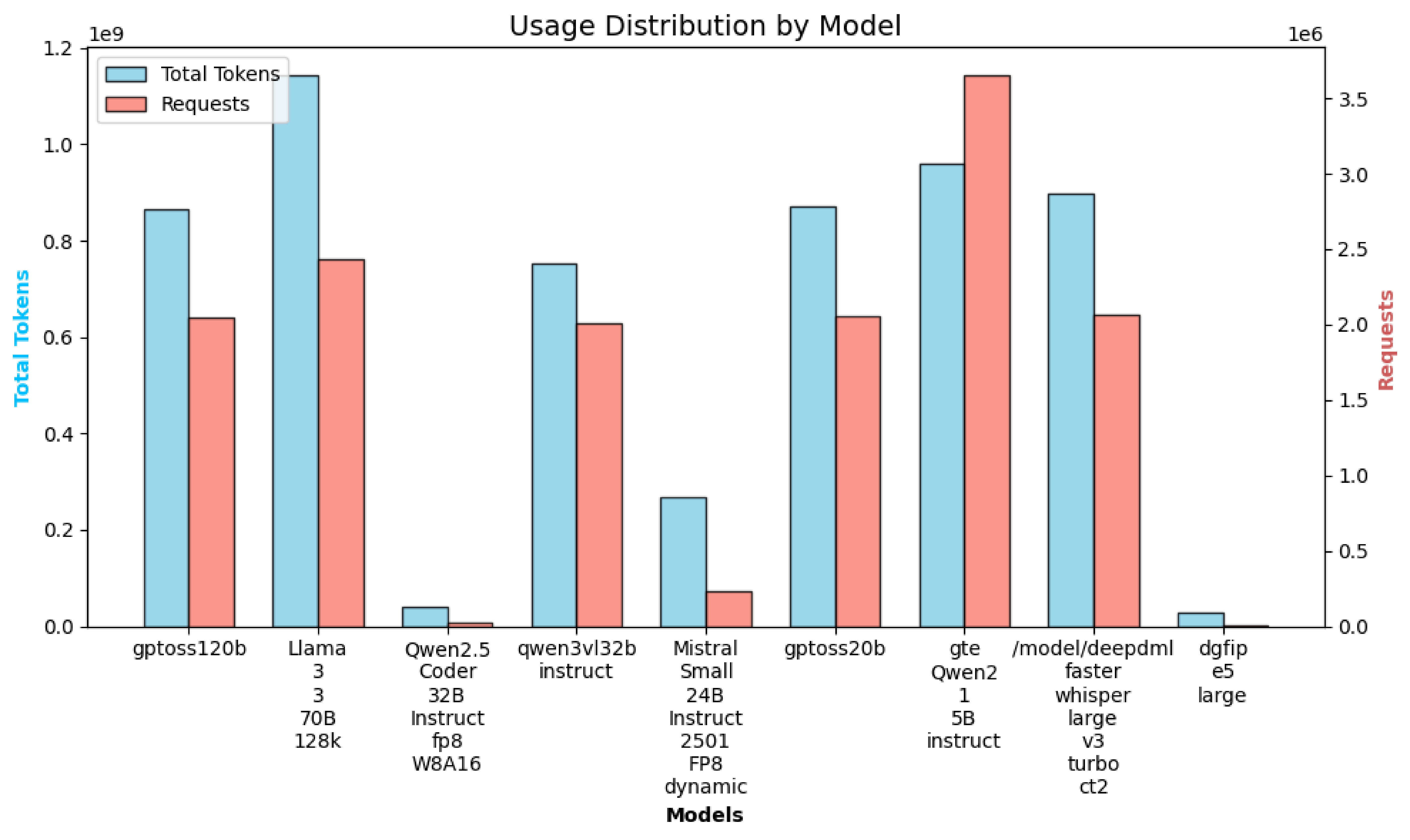


PHASE III: INFERENCE
Descriptive statistics: Usage data



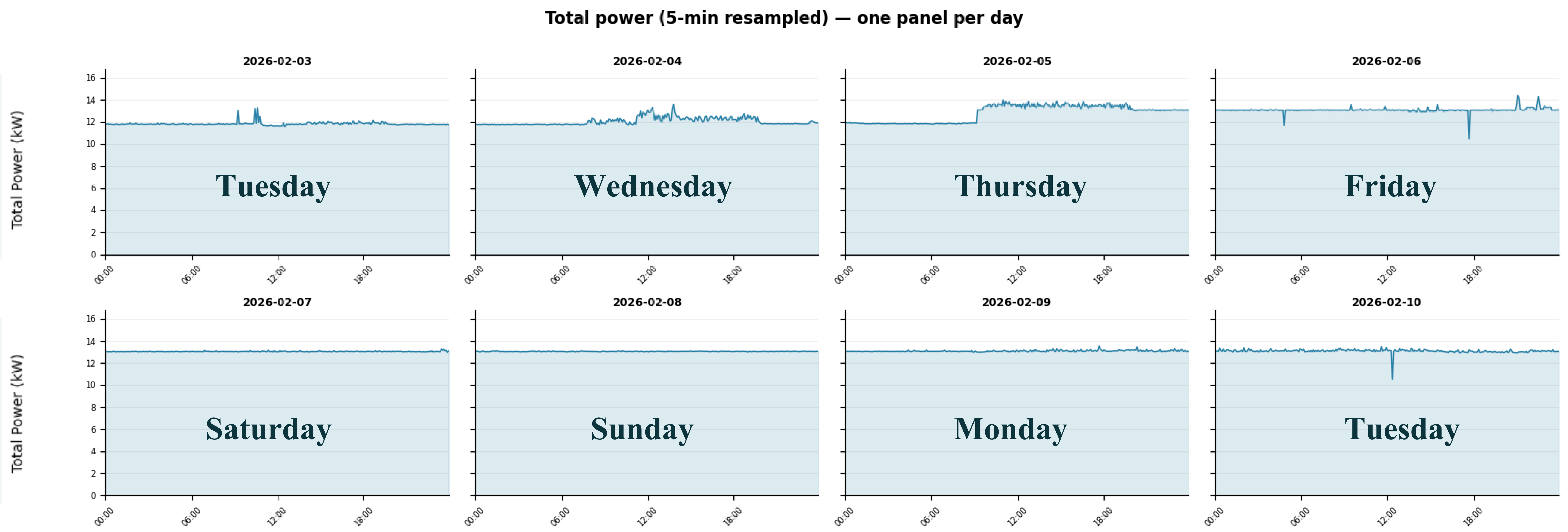
- **Total tokens and requests per day:**
  - **Requests:** in hundred thousands
  - **Tokens:** in hundred millions

PHASE III: INFERENCE
Descriptive statistics: Usage data



PHASE III: INFERENCE
Descriptive statistics: Energy data

Extract of a week of power consumption every 5min from a portion of the 17 days provided



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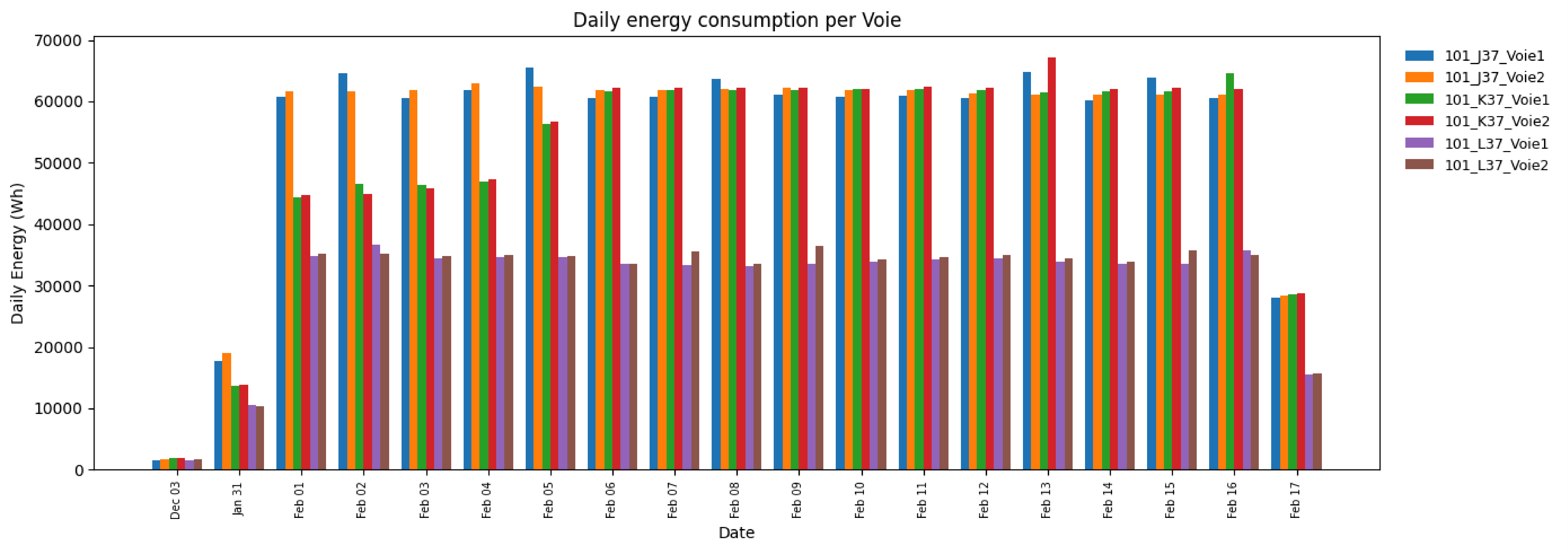


ASSESSMENT TOOL
The methodology

Step 1: Measure total energy

- Power meters (PDUs) on the GPU servers log energy every minute
- We sum all reading for the month, then apply a PUE factor of 1.7

Step 2: Attribute energy across models

- We calculated each model’s share of energy usage as tokens processed by model size, applying three corrections:
  - **Quantisation:** FP8 models use around 50% less compute than BF16 equivalents
  - **MoE architecture:** gptoss models activate only a small fraction of paramenters per request
  - **Idle power:** estimate around 7% of energy draw by GPU just staying on, regardless of usage

ASSESSMENT TOOL

Attribution corrections applied to the energy model

1. Numerical precision (q_i)

Lower precision = fewer FLOPs per token:

Format	q_i	Models
BF16 / FP16	1.00	Llama, qwen3vl, Whisper, gte
FP8	0.50	Qwen-Coder, Mistral
MXFP4-MoE	0.34	gptoss120b, gptoss20b

2. Mixture-of-Experts (MoE)

Only a few experts fire per token -- effective compute << total parameters:

Model	Total params	Active params	Architecture
gptoss120b	120B	15B	128 experts, top-4
gptoss20b	22B	5B	32 experts, top-4

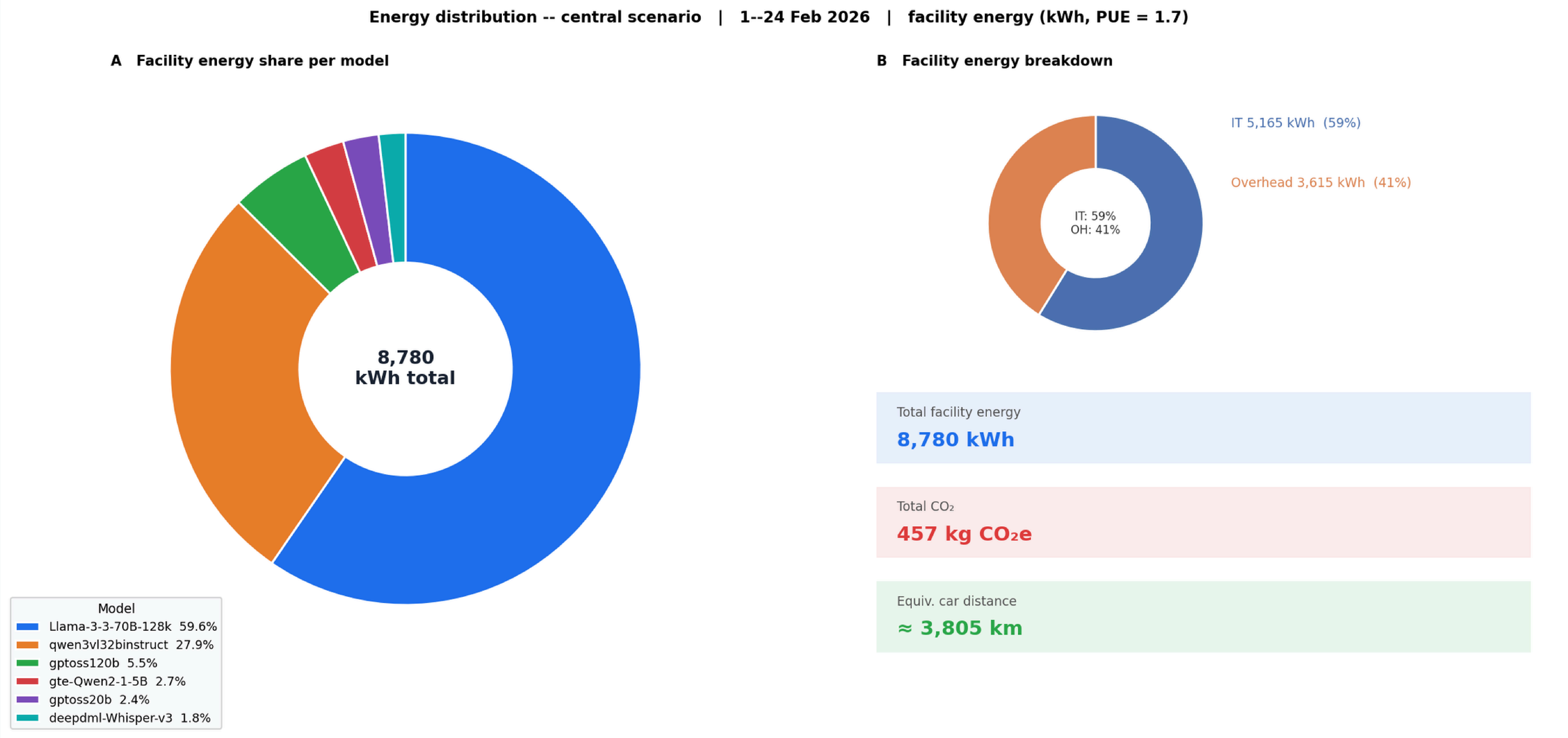
3. GPU idle power

A100 GPUs stay in high-power state (P0) even with no active requests (~50 W idle).

7% of cluster energy is allocated by hosting time (d_i), not by token volume.

$$s_i = 0.93 \cdot \frac{w_i}{\sum_j w_j} + 0.07 \cdot \frac{d_i}{\sum_j d_j}$$

ASSESSMENT TOOL
February 2026 Results



ASSESSMENT TOOL
February 2026 Results

Results per model -- 1--24 Feb 2026 (facility: 8,780 kWh | CO_2: 457 kg | PUE=1.7 | 0.052 kg/kWh)

Model	Share	kWh	CO_2 (kg)	kWh/Mtok	Wh/req	gCO_2/req
Llama-3-3-70B-128k	60%	5,233	272	16.91	4.82	0.251
qwen3vl32binstruct	28%	2,447	127	7.91	2.25	0.117
gptoss120b	6%	484	25	1.57	0.45	0.023
gte-Qwen2-1-5B	3%	240	13	0.78	0.21	0.011
gptoss20b	2%	214	11	0.69	0.20	0.010
deepdml-Whisper-v3	2%	160	8	0.52	0.15	0.008
Total	**100%**	**8,780**	**457**	**4.73**	**1.34**	**0.070**

ASSESSMENT TOOL
Limitations

Assumptions and limitations

- Deploy/undeploy per model
- Which GPU host which model?
- Power per GPU used
- Dec power from 80 min snapshot
- No January 2026 data

Strategic recommendations for DGFiP

- **Implement GPU-level monitoring:** nvidia-smi : System Management Interface:
  - **Power draw** per GPU and not the PDU for the rack
  - **Memory usage:** model deployment management
- **Distinguish between input/output tokens**

MODEL ENERGY MONITORING TOOL

- In order to overcome estimation limitations, we developed a monitoring tool to enable your team to track the energy intensity of each model you deploy.
- The goal of the tool is for your team to automate **the collection, analysis, and integration of the model energy** data in your model choice decision process.
- While the tool currently offers model energy data, it requires development on your end to refine data aggregation, configure to internal data sources, and extend to all models you currently deploy

MODEL ENERGY MONITORING TOOL

- Relying on the February energy and model deployment data, **the tool automatically collects and organizes the usage and deployment figures for each model.** We aim to measure **Wh per 1000 queries**, the industry standard, for each model deployed.

- The energy data provides **cumulative data of Wh per five minute intervals per voie**. The model usage data provides **number of tokens per day,** aggregating all models used in one day to the same token usage.

- With the first two days of February only running one model, we are able to isolate the energy and usage data for the **Qwen 2 Instruct model.** This is not to offer a robust estimate but rather develop the algorithm for your team to run on your side to quickly capture energy estimates for all your models.

Next GTE Qwen 5B Instruct:

- **878.30 Wh / 1000 tokens**
- **30.07 gCO2e / 1000 tokens (8.31 kgCO2e total)**
- **138 meters in thermic car / 1000 tokens**

LIMITATIONS OF OUR STUDY

1 Time Framing

We currently pull the energy intensity of the French energy grid by fifteen minute intervals, the lowest they offer. However, if you decide to deploy models outside of France, this algorithm will underestimate carbon emissions. We opt for **real-time** data, instead of **historical**, so the RTE data only pulls for the past 1.5 months.

2 Estimation Precision

Idle energy of the server can compromise up to 10% of the overall energy of all models, but we do not have data that reveals server downtime. Capturing this will correct for overestimation in energy use and carbon emissions.

3 Data Precision

Model level and intra-day level data is necessary for token and request data. Currently, measurements do not allow us to capture dispersion either between models or within models, besides for one model over two days.

4 Model Characteristics

We encourage you to make use of HuggingFace API to compare the characteristics of models for future deployment, including **ex-post testing** before production deployment and potentially **ex-ante estimation** to minimize energy footprint.

MODEL ENERGY MONITORING TOOL

Strategic recommendations for DGFiP

- Thus, there is no model usage dispersion to exploit to estimate per model usage. **Disaggregated data is needed**:
    - **Total tokens per model per period of time**
    - **Model run times**, preferably at the minute-level. This will reveal when the server is idle.
- Furthermore, data on the GPU where a model is run will improve the reliability of the model energy estimate

PHASE IV: END-OF-LIFE MANAGEMENT

- **The data gap**: Lack of AI-specific e-waste metrics. (OECD 2024)
- **The obsolescence paradox due to rapid gains:** gCO2e/Exaflop improves so fast it incentivises premature hardware replacement before amortisation:
  - Amortisation of classical servers: 5-7 years
  - Amortisation of high-performance GPUs (for AI): 3 years

  (Wang et al., "Computational waste from generative AI" (2024))
- **Dismantling/Transport**: High logistics footprint depending on specific hardware.
- **Closed-loop recycling:** Potential to offset 4% of manufacturing carbon footprint.
  (Schneider et al., "LCA of AI hardware" (2025))

PHASE IV: END-OF-LIFE MANAGEMENT

Strategic recommendations for DGFiP

- Extend the lifespan of GPU by 1 year
- Work with Dell on industy standard recycle methodology
- **NumEcoEval:** developed by CGDD. to quantify the environmental footprint of every hardware component and digital service within the organization

(https://lesbases.anct.gouv.fr/ressources/numecoeval-calculer-l-empreinte-environnementale-de-son-systeme-d-information)

CONCLUSION

Hardware and infrastructure: Extend GPU lifespan, improve data center efficieny (PUE cooling)

Initial model training: Prioritise models with transparent environmental reporting. Look at key indicators for environmental impact (model size, training location, dataset volume, hardware used)

Inference: Use the developed assessment tool to estimate inference emissions and compare models. Improve data granularity and deploy GPU-level monitoring to refine estimates.

End of life: Extend hardware lifespan and integrate recycling strategies.

We have created a **Github repository** that centralises all these project materials :

- Comprehensive report detailing each stage of the AI lifecycle assessment (literature and methodology for tool)
- Code for both tools
- Data descripton code

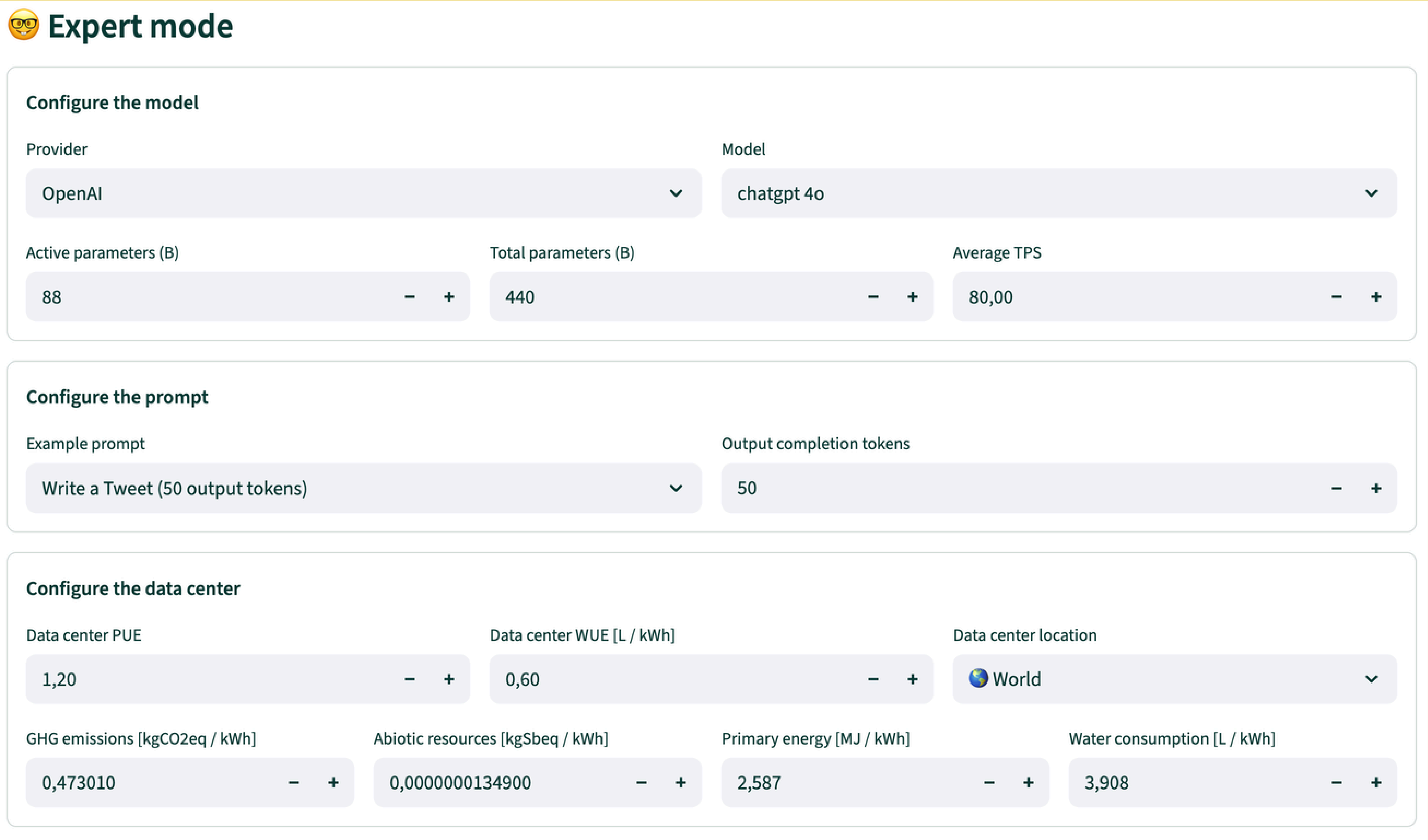
CONCLUSION

Other ways to choose your future models

- **Ecologit carbon is an open source tool for estimating the energy consumption and environmental footprint** when using generative AI models : (https://huggingface.co/spaces/genai-impact/ecologits-calculator)

- **AI energy score :** an initiative to establish comparable energy efficiency ratings for AI models. They use the same chips as DGFiP (H100 GPUs), ideal to make comparisons across wide model types.

(https://huggingface.co/spaces/AIEnergyScore/Leaderboard)



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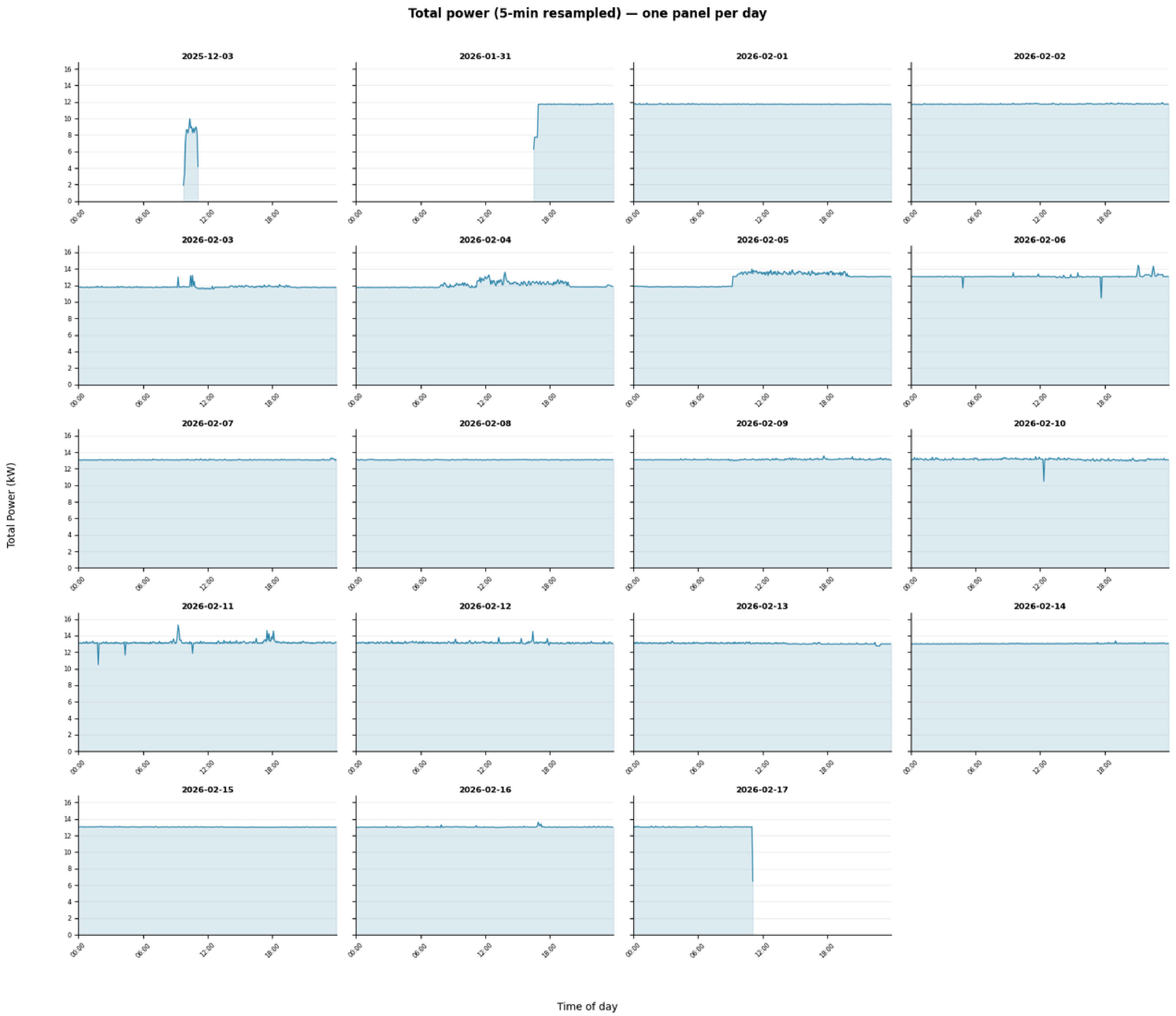
REFERENCES

The full list is in the report

- Berthelot, Adrien, Eddy Caron, Mathilde Jay, and Laurent Lefèvre. 2024. **"Estimating the Environmental Impact of Generative-AI Services Using an LCA-Based Methodology."** Procedia CIRP 122: 707–712.
- Ciambrone, David F. **Environmental Life Cycle Analysis.** 1st ed. Boca Raton, FL: CRC Press, 1997.
- Eilam, Tamar, Pedro D. Bello-Maldonado, Bishwaranjan Bhattacharjee, Carlos Costa, Eun K. Lee, and Asser Tantawi. T**owards a Methodology and Framework for AI Sustainability Metrics.** IBM T. J. Watson Research Center, Yorktown Heights, USA.
- Mistral AI, **"Our Contribution to a Global Environmental Standard for AI,"** July 22, 2025
- Ligozat A.-L., Lefevre J., Bugeau A. & Combaz J. 2022. **Unraveling the hidden environmental impacts of AI solutions for environment life cycle assessment of AI solutions.** Sustainability, 14(9):5172, DOI: 10.3390/su14095172
- Luccioni A.S., Viguier S. & Ligozat A.L. 2023. **Estimating the carbon footprint of BLOOM, a 176B parameter language model**. Journal of Machine Learning Research, 24(253):1−15, https://jmlr.org/papers/v24/23-0069.html
- Ludvigsen, Kasper Groes Albin. **"Carbon Footprint of LLM Fine Tuning — A Case Study."** Towards Data Science, Medium, February 22, 2024.
- Nestor Maslej et al., Artificial Intelligence Index Report 2025, arXiv preprint arXiv:2504.07139v3 (2025), https://doi.org/10.48550/arXiv.2504.07139
- OECD. **Measuring the Environmental Impacts of Artificial Intelligence Compute and Applications: The AI Footprint.** OECD Digital Economy Papers, No. 341. Paris: OECD Publishing, November 2022.
- Plociennik, Christiane, Ponnapat Watjanatepin, Karel Van Acker, and Martin Ruskowski. 2025. **"Life Cycle Assessment of Artificial Intelligence Applications: Research Gaps and Opportunities."** Procedia CIRP 135: 924–929.
- Touvron, Hugo, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, et al. 2023. **"Llama 2: Open Foundation and Fine-Tuned Chat Models."** arXiv preprint arXiv:2307.09288v2.

Annex

Power consumption per day over the full sample of days availables



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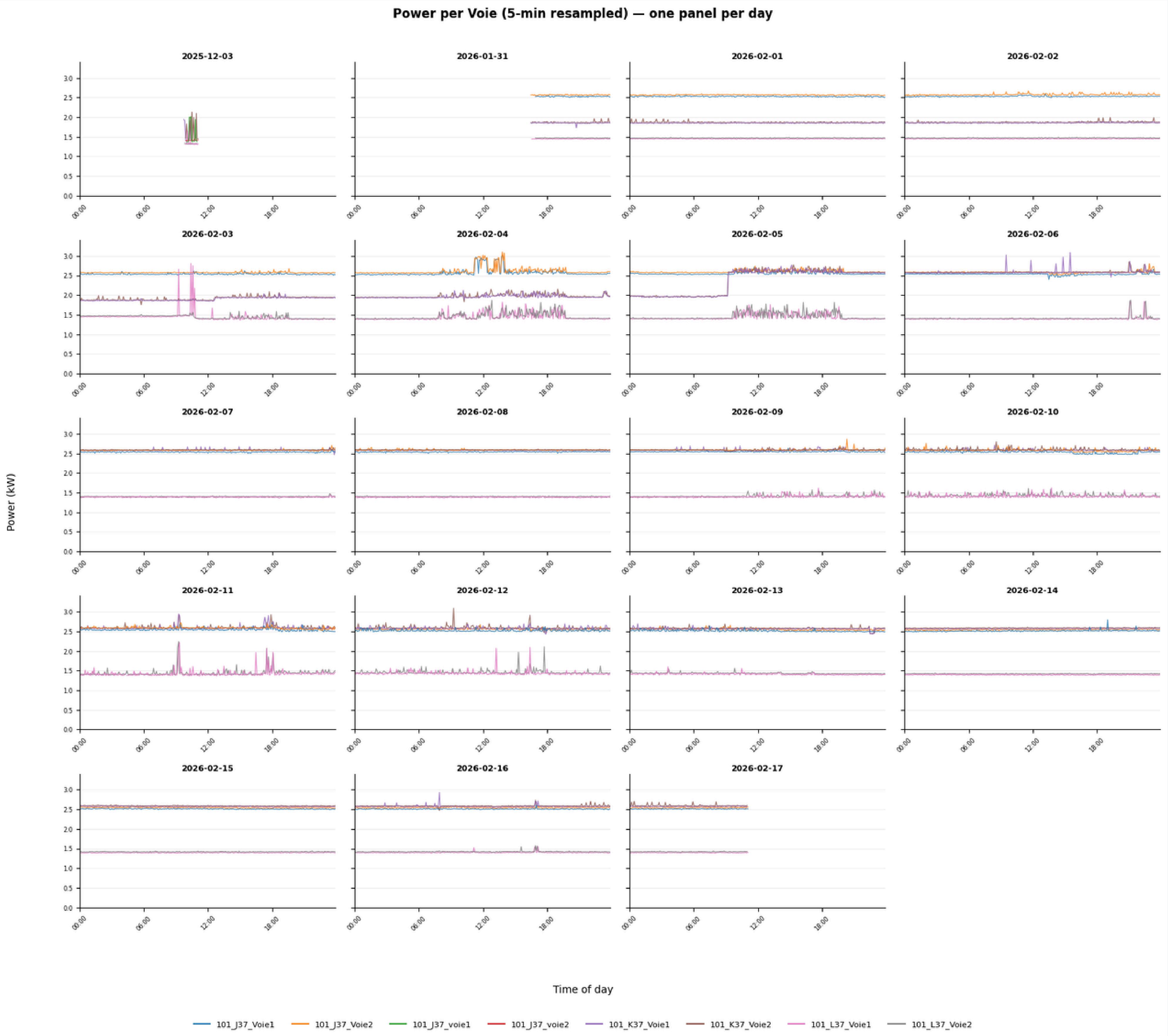
Power consumption per day per Voie



Annex

Average power consumption per Voie